SUPERNOVA × SPACEPOINT · THEME 03 · SEAS OF DATA

Radar sees every ship. We find the ones that hide.



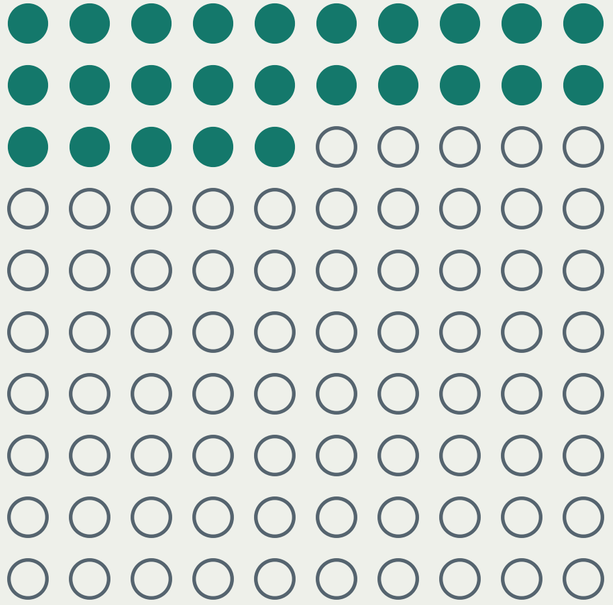
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BITS Pilani, Dubai Campus · November 2026

About three in four industrial fishing vessels are missing from public ship tracking

72–76%

of the world’s industrial fishing vessels are not publicly tracked. AIS is a signal each ship chooses to send.

Global figure, 2017 to 2021, not specific to the UAE. Dark does not mean illegal, and radar shows that a ship is there, not who it is.



Each circle = 1 of 100 vessels.
Filled = tracked.

SOURCE: PAOLO ET AL., NATURE 625, 85-91 (2024)

Sentinel-1 passes over Jebel Ali about every three days

COVERAGE

250 km swath, 10 m pixels, VV band

CONDITIONS

Day, night and through cloud

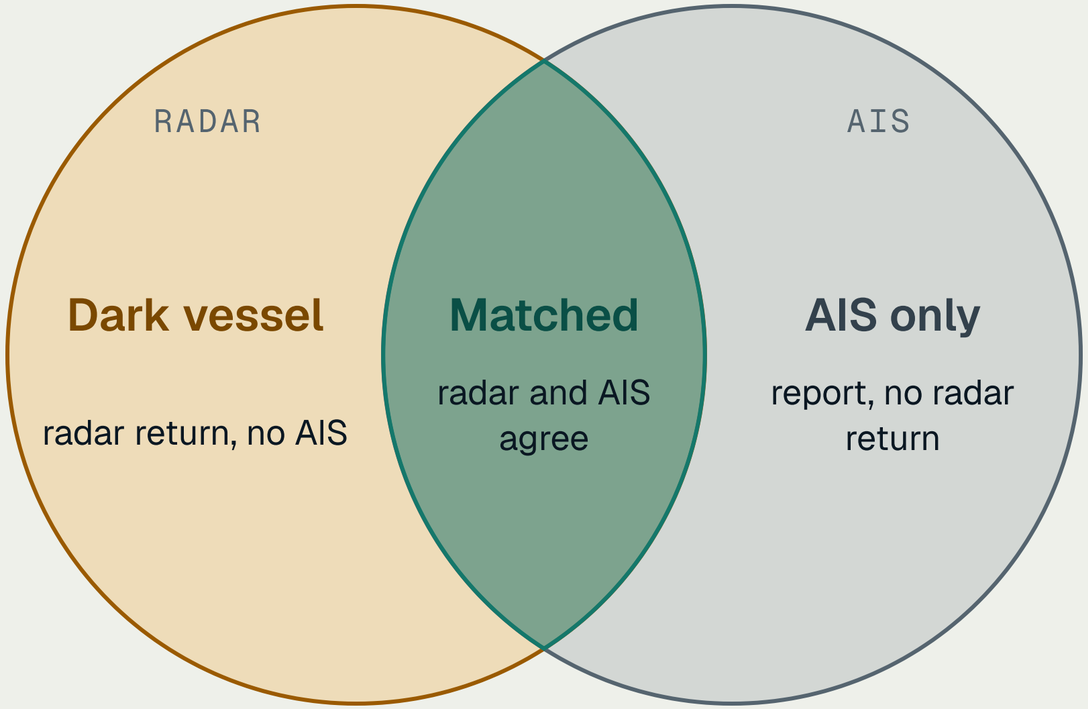
ACCESS

Free and open through Copernicus



PASSES FOUND IN THE COPERNICUS CATALOGUE · 25.01°N 55.06°E

Radar minus AIS equals the dark vessels



THE RULE

A ship is matched when an AIS report lies within 500 m of a radar detection. No report in range: it is flagged dark.

WHY IT HELPS

Dark is a lead, not a verdict. It tells patrols and port authorities where to look first.

IN THIS PROTOTYPE

AIS reports are simulated, because no free real Gulf feed exists. Every output is labelled that way.

Four real Gulf passes: 250 ships located and checked against a simulated AIS list



2 NOV 2026 · 02:14 UTC · 20 OF 73 SHIPS SHOWN

250
SHIPS

205
MATCHED

45
DARK

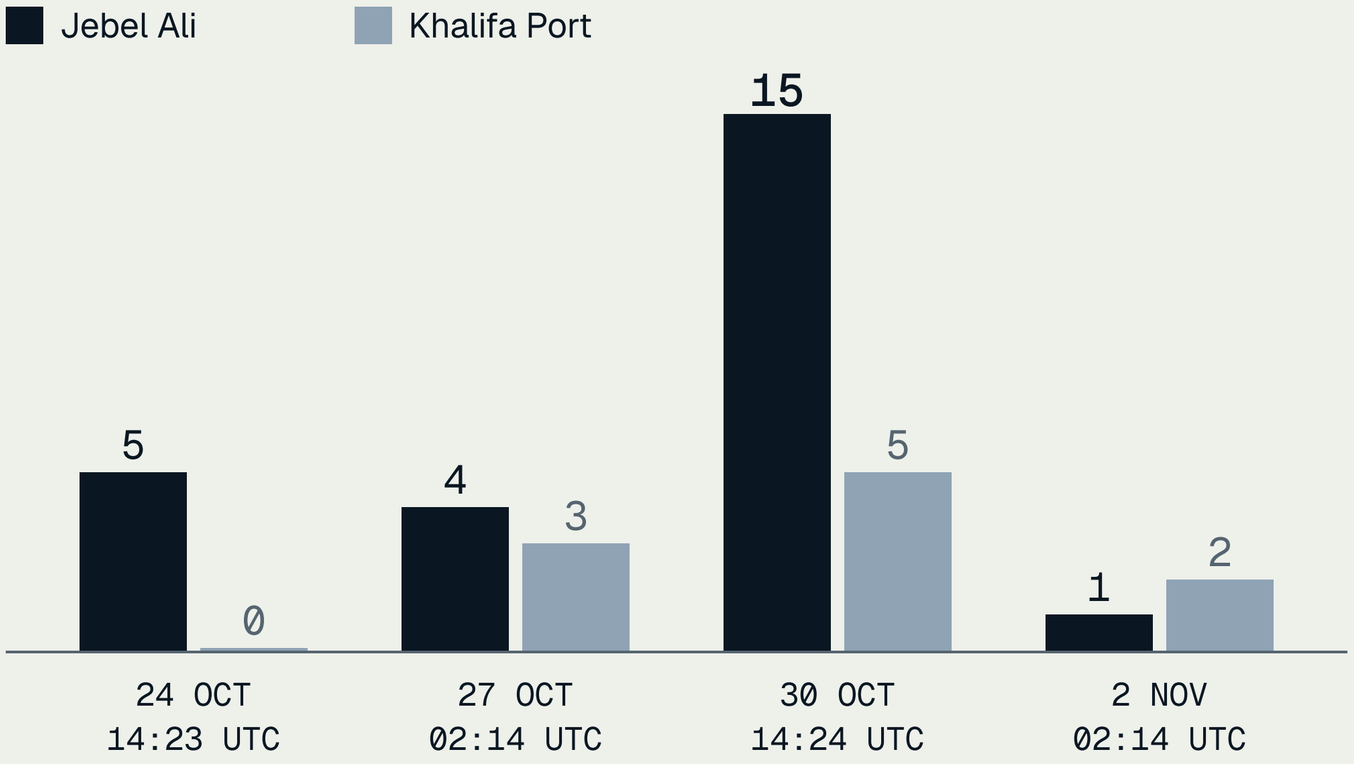
AIS IS SIMULATED

The dark share follows from how we simulate AIS, so it shows the method working, not a finding about Gulf shipping.

HOW TO READ IT

Teal ring: an AIS report within 500 m. Amber ring: none. Each amber ring is a lead to check, not an accusation.

Four passes give a baseline, not a trend: Jebel Ali’s approaches held 1 to 15 ships



THE INDEX

Ships counted in each port box on a pass, divided by that port’s median across passes, times 100.

WHAT THE COUNTS ARE

Ships at anchor and in the approaches. Ships hidden among quay clutter are mostly not counted.

WHY NO TREND YET

Sea state and look angle change what radar picks up, and we have not corrected for them.

SHIPS DETECTED PER PORT BOX · 4 PASSES

On 1,962 radar images it never saw, the detector finds 82% of ships at 91% precision

5,604 HRSID IMAGES, SPLIT THREE WAYS



3,278 train

364 tune

1,962 never seen



Scores on the held-out test split. Bars run from 0 to 1.

WHY IT MATTERS

A dark flag is only as trustworthy as the detection under it. Misses hide ships and false alarms invent them.

WHAT THIS IS NOT

HRSID pixels are finer than free Sentinel-1's 10 m, and the Gulf scenes have no labels, so Gulf accuracy is not measured.

WHAT WE DID ABOUT IT

The Gulf passes use a classical CFAR detector that needs no training. Appendix A2 has the settings.

What is simulated, what is assumed, and what comes next

Item	Status	Effect on the result
AIS reports	Simulated	Dark share is illustrative until a real feed is connected
Port capacity	Assumed	Scales berth occupancy only; the activity index is unaffected
Training data	Public HRSID	Finer pixels than Sentinel-1, so Gulf passes use a classical detector
Buoy	Design only	CAD model and renders, nothing built

NEXT

1. Connect a real AIS feed
2. Retrain on xView3-SAR, which labels dark vessels
3. Add oil-spill detection on the same radar scenes

Data, models and licences

Component	Used for	Source and terms
Sentinel-1 GRD	Real Gulf scenes, VV band	ESA, via Copernicus Data Space, free and open
HRSID	Training and held-out test of the ship detector	Wei et al., IEEE Access; repository under GPL-3.0
YOLO11	Detector, fine-tuned from pretrained weights	Ultralytics
Port boxes	Counting ships inside Jebel Ali and Khalifa Port	Drawn by us, checked against OpenStreetMap extents

Contains modified Copernicus Sentinel data 2026.

Method and parameters

Step	Setting	Reason
Sea mask	Static lon/lat grid, 0.001°, built from a calm-sea pass	Windy sea is as bright as desert, so brightness cannot find the coast
Detector	CFAR, 61 px window, 8 sigma, at least 3 px, at most 450 m	Needs no training and works at 10 m pixels
Merge	Returns within 150 m become one ship	One hull often splits into several blobs
Geocoding	Ground control points in each GRD file	Checked against Palm Jumeirah and Palm Jebel Ali
AIS match	Simulated: 80% reported, 120 m jitter; gate 500 m	No free real Gulf AIS feed
YOLO11s	Fine-tuned on HRSID, scored on 1,962 held-out images: precision 0.91, recall 0.82, mAP50 0.90	Benchmark score only; Gulf passes use CFAR